

DevLabs AI Agent Architecture

Builder & Founder portals — implementation reference

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Short answer: one agent or two?

Two separate agents, not one shared agent. They share infrastructure (OpenRouter, actions API, Supermemory, MongoDB) but have different system prompts, tools, and behavior.

Agent	API action	Persona
Builder Agent	builder_chat	Profile intelligence — fix proof, imports, availability
Founder Agent	founder_chat	Hiring — role briefs, search, shortlist, intros

There is also a **legacy** founder parser in `founderAgent.ts` (`parseFounderAgentTurn`) that is **not wired to the UI** anymore; the live founder flow uses `founder_chat` only.

Architecture (high level)

```
UI Layer BuilderAgentTab / BuilderOSDashboard → builder_chat
FounderRoleIntakeChat / Onboarding → founder_chat
Other dashboards (intros, trials...) → 40+ discrete actions ↓ POST /api/agent/actions
(src/pages/api/agent/actions.ts) ↓ actionsHandler.postAgentAction
(src/lib/agent/actionsHandler.ts) ↓ builder_chat / founder_chat →
generateOpenRouterAgentTurn (tool loop)
other actions → generateOpenRouterReply
or deterministic code ↓ OpenRouter API (src/lib/openrouter.ts) ↓ MongoDB +
optional Supermemory (src/lib/talent/supermemory.ts)
```

Entry point: All agent traffic is `POST /api/agent/actions` with body `{ action, payload }`, authenticated via JWT cookie/header. May proxy to a Node backend when `shouldUseApiProxy` is set.

LLM layer (src/lib/openrouter.ts)

Environment variable	Role
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<code>OPENROUTER_API_KEY</code>	Required for AI; without it, chat degrades / fallbacks
<code>OPENROUTER_MODEL_CHAT</code>	Default: <code>google/gemini-2.5-flash</code>
<code>OPENROUTER_HTTP_REFERER</code>	Optional OpenRouter header
<code>OPENROUTER_APP_NAME</code>	Optional X-Title header
<code>SUPERMEMORY_API_KEY</code>	Optional cross-session memory

Two call styles

- 1. **generateOpenRouterReply** — single turn, no tools (legacy parse, Devpost scrape, getMessage).
- 2. **generateOpenRouterAgentTurn** — OpenAI-style tool calling (`tools` , `tool_choice: auto`).

Function	Default temperature	Default max_tokens
generateOpenRouterReply	0.2	400
generateOpenRouterAgentTurn	0.3	600 (overridden per call)

Builder agent (builder_chat)

Where it is used

- `BuilderOSDashboard.tsx` — handleAgentSend → builder_chat (+ resume upload → chat again)
- `BuilderAgentTab.tsx` — Profile Agent tab via BuilderDashboardContext
- `BuilderHomeTab.tsx` — nudges user toward agent tab

Request payload

```
{
  "action": "builder_chat",
  "payload": {
    "message": "user text",
    "history": [{ "role": "user" | "assistant", "content": "..."}]
```

```
}  
}
```

Agentic loop

- Calls `generateOpenRouterAgentTurn` with tools
- **MAX_ITERATIONS = 5**
- **temperature: 0.3, maxTokens: 600**
- Tools executed in parallel per turn; results appended as `role: tool` messages

Builder tools (11)

Tool	Purpose
<code>get_builder_profile</code>	Read profile, scores, quality issues
<code>get_projects</code>	List projects
<code>update_availability</code>	availableNow, hoursPerWeek, remotePreference
<code>update_links</code>	GitHub, LinkedIn, portfolio
<code>update_profile_basics</code>	headline, bio
<code>import_project</code>	Devpost/GitHub scrape → ProjectRecord
<code>evaluate_profile</code>	Run quality eval, save to profile
<code>fetch_intros</code>	Intro inbox
<code>fetch_work_trials</code>	Active trials
<code>fetch_calls</code>	Upcoming calls
<code>fetch_messages</code>	Threads or single thread history

Memory: `getBuilderMemoryProfile(builderId, userText)` injected into system prompt; writes via `addBuilderMemory` on updates/imports.

Response shape

```
{  
  "success": true,  
  "message": "final assistant text",  
}
```

```

"uiBlocks": [{ "type": "profile_quality" | "summary_card", ... }],
"builder": { /* refreshed profile */ },
"meta": { "model": "...", "iterations": N }
}

```

Founder agent (founder_chat)

Where it is used

- `FounderRoleIntakeChat.tsx` — main chat; sends synthetic `__init__` on first load
- `FounderOnboardingChat.tsx` — wrapper around `FounderRoleIntakeChat`
- `FounderOSDashboard.tsx` — embeds intake when no roles / editing

Non-chat founder UI still calls `/api/agent/actions` with actions like `run_builder_search`, `request_intro`, `generate_trial_project`, etc.

Request payload

```

{
  "action": "founder_chat",
  "payload": {
    "message": "user text or __init__ synthetic",
    "opportunityId": "optional Mongo id",
    "history": [ /* last ~16 turns */ ]
  }
}

```

Server loads current opportunity (by opportunityId or latest draft), founder startup context, and Supermemory founder profile.

Founder tools (8)

Tool	Purpose
<code>get_all_roles</code>	List founder opportunities
<code>get_role_brief</code>	Current brief + missing fields
<code>create_role_brief</code>	New Opportunity (sanitized)
<code>update_role_brief</code>	Patch brief + merge hints from user text

run_search	Rank builders, upsert MatchRecord + Shortlist
get_shortlist	Anonymous or unlocked candidates
get_candidate_details	Full candidate + projects
request_intro	Creates IntroRequest, notifies builder

- Loop: max 5 iterations, temperature 0.3, maxTokens 700
- UI blocks from `buildFounderUiBlocks` in `founderAgent.ts`
- Frontend parses numbered options for option cards

Response shape

```
{
  "success": true,
  "message": "...",
  "opportunity": { /* Opportunity doc */ },
  "uiBlocks": [...],
  "session": {
    "currentSearchId": "...",
    "currentRoleBrief": {...},
    "skippedFields": [],
    "missingRequiredFields": [],
    "shortlistState": "none" | "locked" | "unlocked"
  },
  "meta": { "model": "...", "iterations": N }
}
```

Legacy founder parser (not live chat)

`src/lib/talent/founderAgent.ts` defines `parseFounderAgentTurn`: one-shot JSON extraction (intent, message, extractedData). **Not imported by UI.** Only `buildFounderUiBlocks` is used from that module in `actionsHandler`.

Agent vs actions API

`postAgentAction` is a large RPC router (~30+ actions). Only two are full agentic tool loops: `builder_chat` and `founder_chat`.

Category	Example actions
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Builder profile	claim_profile, update_availability, update_links, evaluate_profile_quality, get_builder_dashboard
Founder hiring	create_role_brief, update_role_brief, run_builder_search, unlock_shortlist, founder_candidate_action
Intros / calls / trials / messages	request_intro, schedule_call, send_trial_project, get_thread_messages, ...
Notifications	get_notifications, mark_notification_read

`ConversationalShell.tsx` is an older demo UI calling discrete actions, not `*_chat`.

Other LLM touchpoints (not full agents)

- `Devpost import` inside `scrapeAndImportProject` — extract JSON from markdown
- `getAgentMessage` — polished copy after successful mutations
- `founderTrialProject.ts` — generate trial project copy
- `roleBriefFieldEnhance.ts` — enhance a single brief field
- `routeProjectCrudWithLLM` — defined but not called (dead path)

Auth and data flow

1. JWT from cookie/Authorization → `resolveAuthedBuilder` or `resolveAuthedFounder`
2. Tools read/write Mongo models (`BuilderProfile`, `Opportunity`, `Shortlist`, `MatchRecord`, ...)
3. Client refreshes local state from response (`builder`, `opportunity`, `uiBlocks`)

File map

File	Role
<code>src/lib/openrouter.ts</code>	OpenRouter client, tool types
<code>src/lib/agent/actionsHandler.ts</code>	All actions + both agent loops
<code>src/pages/api/agent/actions.ts</code>	HTTP entry
<code>src/lib/talent/supermemory.ts</code>	Optional memory

src/lib/talent/founderAgent.ts	UI blocks + legacy parseFounderAgentTurn
src/components/builder/BuilderOSDashboard.tsx	Builder chat client
src/components/founder/FounderRoleIntakeChat.tsx	Founder chat client

Mental model

- **Two conversational agents** (builder vs founder), each with its own prompt and tools.
- **One gateway** (/api/agent/actions) for chat and non-chat operations.
- **OpenRouter** is the only LLM provider for agents; model is env-driven.
- **Supermemory** is optional long-term memory.
- **Not** a single shared agent class — different tools and semantics by role.